

SOFTWARE REQUIREMENTS SPECIFICATION

Bridging Healthcare Gaps with AI

Project Code: BHG-AI-(25-26)
Version: 2.0 (Revised for 4-Month Timeline)
Issue Date: November 19, 2025
Status: Ready for Development

DOCUMENT INFORMATION

Item	Details
Customer	Patients, Healthcare Providers (Doctors)
Project	HealthBridge.ai - AI-Powered Healthcare Platform
Document	Software Requirement Specifications
Version	2.0
Identifier	BHG-AI-(25-26)
Team Size	4 Final Year CS Students
Timeline	4 Months
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DEFINITIONS, ACRONYMS & ABBREVIATIONS

Term	Description
API	Application Programming Interface
JWT	JSON Web Tokens
NLP	Natural Language Processing
OCR	Optical Character Recognition
WebRTC	Web Real-Time Communication
HTTPS	Hypertext Transfer Protocol Secure
MVP	Minimum Viable Product
SRS	Software Requirements Specification

Term	Description
UI/UX	User Interface / User Experience
SMS	Short Message Service
Firebase	Google's Backend-as-a-Service Platform
Agora	Third-party Real-time Communication Platform
PMC	Pakistan Medical Commission

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1. INTRODUCTION

1.1 Purpose of Document

This Software Requirements Specification (SRS) document defines all functional and non-functional requirements for HealthBridge.ai, an AI-powered telemedicine platform designed to improve healthcare accessibility in Pakistan. This document serves as a contract between the development team and stakeholders, outlining what the system will deliver within a 4-month academic timeline with a focus on a Minimum Viable Product (MVP) approach.

1.2 Project Vision

HealthBridge.ai addresses critical healthcare gaps in Pakistan by providing:

- Instant AI-powered health guidance through conversational chat

- Automated interpretation of medical reports in plain language
- Real-time teleconsultation with licensed doctors
- Intelligent doctor matching based on specialty and location
- Support in English and Urdu languages

The platform bridges geographical and financial barriers to healthcare, particularly benefiting rural and underserved populations who lack access to specialist doctors and timely medical consultations.

1.3 Problem Statement

Current Challenges in Pakistan's Healthcare:

- 50% of the population lacks access to primary healthcare services
- Rural areas face emergency response times averaging 14+ minutes
- Patients cannot access specialist doctors without traveling long distances
- Complex medical reports confuse patients due to use of technical terminology
- Language barriers prevent non-English speakers from seeking online healthcare
- Long appointment waiting times delay urgent medical consultations

Existing Solutions' Limitations: Platforms like Oladoc, Marham, and Sehat Kahani provide appointment booking and basic telemedicine but lack AI-powered preliminary guidance. Patients still must wait for doctor availability and cannot get instant answers to urgent health questions.

1.4 Proposed Solution

HealthBridge.ai will offer:

1. **AI Health Assistant** - Real-time symptom assessment and health guidance
2. **Medical Report Analysis** - Automatic interpretation with plain language explanations
3. **Doctor Teleconsultation** - Video/audio/text real-time consultations
4. **Intelligent Matching** - Connect patients with appropriate specialists
5. **Multi-Language Support** - Full functionality in English and Urdu

2. PROJECT OVERVIEW & SCOPE

2.1 Project Scope - Phase 1 (MVP - 4 Months)

In-Scope Deliverables:

A. AI Health Assistant

- Chat-based interface for health-related queries (English & Urdu)
- Symptom assessment using pre-trained medical NLP models
- Urgency classification (Green: Self-care / Yellow: Consult Soon / Red: Urgent / Emergency)
- Personalized responses based on user medical history
- Confidence scoring for AI responses
- Context-aware conversation with memory (within session)

B. Medical Report Analysis

- Upload medical reports in JPG/PNG/PDF formats
- Optical Character Recognition (OCR) text extraction
- Identification of test values and comparison with normal ranges
- Plain language explanations of abnormal results
- Visual alerts for critical values

C. Doctor Consultation System

- Real-time video calling (using WebRTC/Agora)
- Text-based chat during consultations
- Doctor profile database with specialties and credentials
- Appointment booking with calendar integration
- Doctor matching algorithm (specialty + location-based)
- Appointment reminders via SMS/push notification
- Basic payment gateway integration (JazzCash/EasyPaisa)

D. Voice Agent for Appointment Booking

- Outbound voice calling to clinics for automated appointment scheduling
- Natural language conversation with clinic staff using voice
- Automatic extraction of clinic availability
- Voice-based appointment confirmation and rescheduling

- Fallback to SMS/WhatsApp if voice call fails
- Call transcription and logging for patient records
- Support for both English and Urdu voice interactions

E. User Management

- User registration (as Patient or Doctor)
- Secure login with JWT authentication
- Basic profile management and medical history tracking
- Password recovery functionality
- Session management with timeouts

F. Multi-Language Support

- Full UI in English and Urdu
- Language-switching interface
- AI responses translated to selected language
- Medical terminology accurately translated
- Voice agent supports both English and Urdu conversations

G. Health Tracking (Basic)

- Manual entry of vital signs (BP, glucose, weight, temperature)
- Visual trend display via charts
- Alerts for abnormal vital sign values

2.2 Out-of-Scope (Deferred to Phase 2)

The following features are **deferred to Phase 2** or removed due to 4-month timeline constraints:

Feature	Reason for Deferral
AI-Based Medication Recommendations	Requires medical licensing review; complex regulatory compliance
Wearable Device Integration	Requires proprietary APIs; not essential for MVP
Hospital/EHR System Integration	Requires B2B partnerships; standalone system sufficient for MVP
Insurance Integration	Complex backend; payment via direct methods sufficient
Automated Appointment Calling	Voice API integration complex; manual booking sufficient
Health Forum / Social Features	Low priority for MVP; can be added later
Family Account Management	Adds complexity; single-user accounts sufficient
Advanced Analytics Dashboard	Low priority for core functionality
FDA/HIPAA Certification	Academic prototype; medical disclaimer sufficient

2.3 System Constraints

2.3.1 Time Constraints

- **Development Timeline:** 4 months
- **Team Size:** 4 students with academic coursework commitments
- **Integration Testing:** Limited by availability of real doctors for testing

2.3.2 Technical Constraints

- **Platforms:** Web (modern browsers) + Android mobile app
- **No iOS Support:** iOS development adds complexity; Android sufficient for MVP
- **Internet Dependency:** System requires active internet connection
- **Device Requirements:** Minimum 2GB RAM smartphone/computer

2.3.3 Financial Constraints

- **Third-Party Costs:** API usage fees for Agora, Firebase, OCR services
- **Cloud Hosting:** Limited budget for AWS/Google Cloud
- **Solution:** Use free/affordable tiers during development

2.3.4 Regulatory Constraints

- Must comply with **Pakistan Medical Commission (PMC) guidelines**
- Doctor verification mandatory before account activation

- Medical disclaimers required on all AI recommendations
- Patient data privacy per Pakistan's data protection policies

2.3.5 Resource Constraints

- Limited availability of real doctors for testing phase
 - Restricted access to authentic medical datasets for AI training
 - Pre-trained medical models must be open-source or low-cost
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3. USER CHARACTERISTICS

3.1 Primary Users (Patients)

3.1.1 Rural/Underserved Patients

- Limited access to nearby doctors
- May have limited technical literacy
- Prefer voice support and simple interfaces
- Benefit most from AI preliminary guidance

3.1.2 Urban Patients

- Higher technical literacy
- Prefer quick consultations without travel
- May need specialist doctors frequently
- Use for convenience and time-saving

3.1.3 Elderly Patients

- Require extremely simple, intuitive interfaces
- May need voice guidance through features
- Family members may assist with account setup
- Prefer text/video consultations over travel

3.1.4 Non-English Speakers

- Urdu as primary language

- Require full system support in local language
- May struggle with medical terminology

3.2 Secondary Users (Healthcare Providers)

3.2.1 General Practitioners (GPs)

- Handle routine patient queries
- Create professional profiles with qualifications
- Conduct video consultations during fixed hours
- View patient history during consultations

3.2.2 Specialist Doctors

- Accept consultation requests in their specialty
- Set availability schedules
- Receive AI-generated patient summaries for context

3.2.3 Healthcare Workers

- View basic patient records
- Refer patients to appropriate doctors
- Limited administrative access

3.3 Occasional Users

3.3.1 Family Members

- View patient's basic health information (with permission)
- Assist elderly patients with account management
- Receive appointment reminders

4. SYSTEM ARCHITECTURE & OPERATING ENVIRONMENT

4.1 Hardware Requirements

4.1.1 Server Infrastructure

- **Cloud Hosting:** AWS or Google Cloud Platform

- **Auto-scaling:** Handles concurrent users (estimated 100-500 concurrent users for MVP)
- **Database Server:** Managed database service (AWS RDS / Firebase)
- **Storage:** Cloud storage for medical reports (AWS S3 / Google Cloud Storage)

4.1.2 User Devices

- **Mobile:** Android phones (version 8.0+)
- **Desktop:** Laptops/computers with modern browsers
- **Minimum Specs:** 2GB RAM, stable internet connection
- **Camera/Microphone:** Required for video consultations

4.2 Software Architecture

4.2.1 Web Application

- **Framework:** React.js / Vue.js (for UI responsiveness)
- **Backend:** Node.js + Express or Python Flask/Django
- **Supported Browsers:** Chrome, Firefox, Safari, Edge (latest versions)
- **Responsive Design:** Desktop, tablet, and mobile-friendly

4.2.2 Mobile Application

- **Platform:** Android (React Native or native Android)
- **Minimum OS Version:** Android 8.0+
- **Features:** Full feature parity with web app
- **Offline Capability:** Cache basic features; syncs when online

4.2.3 AI Components

- **NLP Models:** Fine-tuned medical NLP models for symptom analysis
- **Models Used:** Pre-trained BioBERT or MedBERT for symptom analysis
- **Language Support:** English + Urdu (using translation APIs or multilingual models)
- **Deployment:** Cloud-based inference (AWS SageMaker or similar)

4.2.4 Voice Agent Components

- **VoIP Service:** Twilio for outbound calling to clinics

- **Speech Recognition:** Google Cloud Speech-to-Text or Azure Speech Services (real-time STT)
- **Speech Synthesis:** Google Cloud Text-to-Speech (TTS for natural voice responses)
- **Conversation Logic:** Python/Node.js-based conversational AI engine with state management
- **Language Models:** GPT-based or transformer-based models for natural dialogue
- **Call Handling:** Recording, transcription, and logging of all voice calls
- **Fallback Mechanism:** Automatic escalation to SMS or manual booking

4.2.5 Database

- **Primary Database:** PostgreSQL (relational data)
- **Real-time Features:** Firebase Firestore for chat/notifications
- **Data Storage:** User profiles, medical history, consultation records, voice call logs, appointment history

4.3 Network & Communication

4.3.1 Real-Time Communication

- **Video Calling:** Agora SDK (WebRTC-based, supports multiple platforms)
- **Chat:** WebSocket for text messaging during consultations
- **Fallback:** SMS-based notifications for appointments

4.3.2 Network Requirements

- **Connection Types:** 3G/4G/5G mobile networks or WiFi
- **Bandwidth:** Video calling optimized for low-bandwidth scenarios
- **Protocol:** HTTPS for all data transmission; encrypted WebSocket for real-time chat

4.3.3 Security

- **Authentication:** JWT-based token authentication
 - **Encryption:** All data transmitted via HTTPS; sensitive data encrypted at rest
 - **Session Management:** Automatic timeout after 30 minutes of inactivity
 - **Role-Based Access Control (RBAC):** Patients see only their data; doctors see patients' shared records
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5.1 User Management (FR-1)

FR-1.1: Users shall register with email, password, and basic information (name, age, gender, location).

FR-1.2: Users shall select role during registration: Patient or Doctor (immutable after registration).

FR-1.3: The system shall securely hash passwords and enforce password requirements (minimum 8 characters, alphanumeric).

FR-1.4: Users shall be able to reset forgotten passwords via email verification.

FR-1.5: The system shall maintain JWT tokens with 24-hour expiration for authenticated sessions.

FR-1.6: Users shall be able to update profile information (phone, address, medical history).

FR-1.7: Patient profiles shall include medical history fields: chronic conditions, allergies, past surgeries, current medications (for AI context).

FR-1.8: Doctor profiles shall include: medical license number, specialty, years of experience, clinic address, and availability schedule.

FR-1.9: The system shall verify doctor credentials against PMC database before account activation.

FR-1.10: Users shall have logout functionality that clears session tokens.

5.2 AI Health Assistant (FR-2)

FR-2.1: The system shall provide a chat interface for patients to input health queries in English or Urdu.

FR-2.2: The AI assistant shall process natural language queries using fine-tuned medical NLP models.

FR-2.3: Based on patient input, the system shall classify user query into one of 5 categories: Symptoms, Test Results, Medication Query, General Health, or Other.

FR-2.4: For symptom-related queries, the AI shall perform symptom assessment and generate a preliminary diagnosis list (top 3 possibilities with percentages).

FR-2.5: The system shall classify urgency as: Green (Self-care), Yellow (Consult Soon), Red (Urgent), or Emergency.

FR-2.6: The system shall include confidence scores (0-100%) for AI responses; low confidence (<60%) shall trigger recommendation to consult a doctor.

FR-2.7: The AI shall maintain conversation context within a session (remember previous messages in same chat session).

FR-2.8: All AI responses shall be in plain language, avoiding medical jargon where possible; complex terms shall include explanations.

FR-2.9: The system shall provide a disclaimer: "This is not a substitute for professional medical advice. Consult a doctor for diagnosis and treatment."

FR-2.10: Patients shall be able to share AI responses with doctors via in-app sharing button.

FR-2.11: All health queries and AI responses shall be stored in user's medical history for future reference.

FR-2.12: The system shall support multi-turn conversations with context retention within same session.

5.3 Medical Report Analysis (FR-3)

FR-3.1: Patients shall upload medical reports in JPG, PNG, or PDF formats.

FR-3.2: The system shall use OCR (via Google Vision API or Tesseract) to extract text from report images.

FR-3.3: The system shall identify test names and values from extracted text using medical NLP models.

FR-3.4: The system shall compare identified test values against standard normal ranges (predefined database).

FR-3.5: Abnormal values shall be flagged with visual indicators (red for critical, yellow for warning).

FR-3.6: The system shall generate plain language explanations: "Your blood glucose is 180 mg/dL, which is higher than normal (70-100). This may indicate diabetes risk. Consult a doctor."

FR-3.7: The system shall identify report type (Blood Test, X-Ray Report, Ultrasound, etc.) automatically.

FR-3.8: The system shall provide recommendations: "See a doctor within 2 days" or "Schedule urgent consultation."

FR-3.9: Analyzed reports shall be stored in user's medical history with timestamp.

FR-3.10: Users shall be able to share analyzed report summaries with doctors.

FR-3.11: The system shall handle multi-page PDFs by analyzing all pages.

FR-3.12: For unrecognizable reports, system shall display: "Unable to analyze this report. Please consult your doctor for interpretation."

5.4 Doctor Consultation System (FR-4)

FR-4.1: The system shall maintain a database of registered doctors with profiles (name, specialty, qualifications, location, availability).

FR-4.2: Patients shall be able to search doctors by: specialty, location (within X km), or name.

FR-4.3: The system shall display doctor profiles with: photo, qualifications, years of experience, consultation fee, and availability slots.

FR-4.4: Patients shall book appointments by selecting date and time from doctor's availability calendar.

FR-4.5: Upon booking, system shall send confirmation SMS and push notification to both patient and doctor.

FR-4.6: System shall send appointment reminder: 1 hour before scheduled consultation.

FR-4.7: At scheduled time, both patient and doctor shall see "Join Consultation" button that initiates video call via Agora SDK.

FR-4.8: During video consultation, the system shall display patient's medical history (if patient consents) to doctor.

FR-4.9: Patients shall have option to switch between video, audio, or text-chat during consultation.

FR-4.10: The system shall record consultation duration and calculate fees based on time (e.g., first 15 min = Rs. 500).

FR-4.11: Upon consultation end, system shall display payment screen with total fee and payment options (JazzCash, EasyPaisha).

FR-4.12: After payment, system shall generate consultation summary and receipt.

FR-4.13: Consultation records shall be stored in patient's medical history.

FR-4.14: Doctors shall see list of upcoming appointments with patient names and consultation reasons.

FR-4.15: Doctors shall be able to set their availability schedule (available days/times per week).

5.5 Intelligent Doctor Matching (FR-5)

FR-5.1: When patient books appointment for specific symptoms, system shall suggest 3-5 relevant specialists.

FR-5.2: Matching algorithm shall consider: symptom type → specialty mapping, doctor location (within 10 km), and doctor availability.

FR-5.3: System shall display matched doctors sorted by: specialty match score, distance, and availability.

FR-5.4: Patients may override suggestions and book with any available doctor.

FR-5.5: System shall learn from bookings: if patient with symptom X books doctor with specialty Y, increase this association for future matching.

5.5 Voice Agent for Automated Appointment Booking (FR-5)

FR-5.1: Patients shall be able to request "Call clinic to book appointment" from the app.

FR-5.2: The voice agent shall initiate outbound call to clinic phone number using Twilio or similar VoIP service.

FR-5.3: Voice agent shall use speech-to-text (STT) to convert clinic staff responses to text.

FR-5.4: Voice agent shall use text-to-speech (TTS) to communicate appointment options, patient name, symptoms, and availability in natural English or Urdu.

FR-5.5: Voice agent shall extract key information during call:

- Clinic name and location confirmation
- Doctor name and specialty
- Available appointment slots (date and time)
- Patient name, age, and symptom description (if needed by clinic)

FR-5.6: Voice agent shall confirm appointment details back to clinic staff: "So, appointment for Ahmed on December 25th at 3 PM with Dr. Fatima for knee pain. Is that correct?"

FR-5.7: Upon confirmation, voice agent shall:

- Record appointment in system database
- Generate appointment confirmation SMS to patient
- Save call transcript in patient's medical history
- Log appointment in calendar

FR-5.8: If clinic is busy or unavailable, voice agent shall:

- Ask clinic for callback number or best time to call
- Offer to send SMS booking request instead
- Schedule automatic retry after 1 hour

FR-5.9: If voice call fails due to network, system shall:

- Automatically switch to SMS booking: "Sending appointment request to Dr. Clinic. You'll receive confirmation shortly."
- Allow manual follow-up via WhatsApp or direct calling

FR-5.10: Voice agent shall handle common scenarios:

- Doctor is on leave: "This doctor is unavailable. Can I book with another doctor in same specialty?"
- No availability: "No slots available this week. Can I book for next week?"

- Invalid phone number: "This number is not responding. Try manual booking."

FR-5.11: Patients shall receive appointment summary via SMS/push notification:

- Doctor name, clinic location, date/time
- Patient's queued appointment number
- Reminder sent 1 hour before appointment

FR-5.12: Voice agent call duration shall be logged; system shall estimate call time (e.g., "Call duration: 2 minutes").

FR-5.13: System shall maintain call logs accessible to patient and support team for troubleshooting.

FR-5.14: Voice agent shall support handoff to human agent if call becomes too complex (e.g., special medical requirements).

5.6 Intelligent Doctor Matching (FR-6)

FR-6.1: When patient books appointment for specific symptoms, system shall suggest 3-5 relevant specialists.

FR-6.2: Matching algorithm shall consider: symptom type → specialty mapping, doctor location (within 10 km), and doctor availability.

FR-6.3: System shall display matched doctors sorted by: specialty match score, distance, and availability.

FR-6.4: Patients may override suggestions and book with any available doctor.

FR-6.5: System shall learn from bookings: if patient with symptom X books doctor with specialty Y, increase this association for future matching.

5.6 Health Tracking (FR-6)

FR-6.1: Patients shall be able to manually enter vital signs: Blood Pressure (systolic/diastolic), Glucose, Weight, Temperature, Heart Rate.

FR-6.2: Each vital entry shall include timestamp.

FR-6.3: System shall compare entries against normal ranges and flag abnormal values.

FR-6.4: System shall display health trends via line charts (last 7 days, 30 days, 90 days).

FR-6.5: System shall send alerts if vital value exceeds critical threshold (e.g., BP > 180/120 = High Alert).

FR-6.6: Patients shall be able to download health trend reports as PDF.

FR-6.7: Doctors shall view patient's vital history during consultation (if patient consents).

5.7 Multi-Language Support (FR-7)

FR-7.1: User interface shall be available in English and Urdu.

FR-7.2: Users shall switch language via settings; preference shall persist across sessions.

FR-7.3: All AI responses shall be translated to selected language using Google Translate API or multilingual models.

FR-7.4: Medical terminology shall be accurately translated (e.g., "Blood Glucose" → "خون کی شکر" in Urdu).

FR-7.5: Help documentation and FAQs shall be available in both languages.

FR-7.6: Error messages shall display in user's selected language.

5.8 Notifications (FR-8)

FR-8.1: System shall send appointment reminders 1 hour before consultation via SMS and push notification.

FR-8.2: System shall notify doctor when patient books appointment.

FR-8.3: System shall notify patient when doctor accepts/declines consultation request (if added).

FR-8.4: System shall send payment confirmation and receipt via SMS/email.

FR-8.5: For Red/Emergency urgency AI assessments, system shall send urgent notification encouraging doctor consultation.

6. NON-FUNCTIONAL REQUIREMENTS

6.1 Performance (NFR-1)

NFR-1.1: AI response time shall be ≤ 5 seconds for symptom analysis queries.

NFR-1.2: OCR report analysis shall complete within 10 seconds for single-page reports.

NFR-1.3: Page load time for web/mobile app shall be ≤ 3 seconds on 4G connection.

NFR-1.4: Video call initialization shall occur within 5 seconds.

NFR-1.5: Database queries shall complete within 500ms.

NFR-1.6: System shall support minimum 100 concurrent users with response times maintained.

6.2 Reliability (NFR-2)

NFR-2.1: System availability shall be $\geq 95\%$ (excluding scheduled maintenance).

NFR-2.2: Database backups shall occur daily; recovery time objective (RTO) ≤ 1 hour.

NFR-2.3: System shall implement graceful error handling; users shall see friendly error messages, not technical stack traces.

NFR-2.4: For API failures (e.g., OCR service down), system shall retry up to 3 times before informing user.

NFR-2.5: Medical data shall not be lost; all consultations and reports automatically saved.

6.3 Security (NFR-3)

NFR-3.1: All passwords shall be hashed using bcrypt or Argon2 with salt.

NFR-3.2: All data transmission shall use HTTPS with TLS 1.2 or higher.

NFR-3.3: Authentication tokens (JWT) shall have 24-hour expiration; refresh tokens for extended sessions.

NFR-3.4: Sessions shall timeout after 30 minutes of inactivity.

NFR-3.5: Role-based access control (RBAC): Patients access only their data; doctors access patient data only when consultation is booked.

NFR-3.6: Medical data at rest shall be encrypted using AES-256.

NFR-3.7: Payment information (JazzCash, EasyPaisha credentials) shall not be stored; use payment gateway APIs directly.

NFR-3.8: System shall log all data access; audit logs retained for 90 days.

NFR-3.9: Input validation on all user inputs to prevent SQL injection, XSS, and other attacks.

NFR-3.10: CORS (Cross-Origin Resource Sharing) policies strictly configured to prevent unauthorized API access.

6.4 Usability (NFR-4)

NFR-4.1: UI shall require no training; patients unfamiliar with apps should understand basic operations within 2 minutes.

NFR-4.2: Font size minimum 14pt for readability; high contrast for accessibility (WCAG AA standard).

NFR-4.3: Error messages shall be clear and suggest solutions: Instead of "Error 404", display "Doctor not found in your area. Try expanding search radius."

NFR-4.5: Navigation shall be intuitive with clear menu structures; no more than 3 clicks to reach any feature.

NFR-4.6: Voice support for reading chat responses aloud (text-to-speech) for users with visual impairments.

NFR-4.7: All buttons and interactive elements shall be minimum 48x48px for mobile touch targets.

6.5 Scalability (NFR-5)

NFR-5.1: Architecture shall support horizontal scaling; adding more servers increases capacity.

NFR-5.2: Database shall use indexing on frequently queried fields (doctor specialty, user ID, appointment time).

NFR-5.3: Caching layer (Redis) for frequently accessed data: doctor profiles, normal vital ranges.

NFR-5.4: API rate limiting: 100 requests per user per hour to prevent abuse.

NFR-5.5: Cloud infrastructure auto-scales based on load; increases during peak hours.

6.6 Accuracy (NFR-6)

NFR-6.1: AI symptom assessment accuracy shall be $\geq 75\%$ (validated against medical professional reviews).

NFR-6.2: OCR text extraction accuracy shall be $\geq 85\%$ for printed medical reports.

NFR-6.3: Normal range comparisons for vitals shall match standard medical references.

NFR-6.4: Doctor location matching shall have ≤ 1 km error margin using GPS coordinates.

6.7 Compliance (NFR-7)

NFR-7.1: System shall display medical disclaimer on all AI recommendations: "This AI is not a doctor. Always consult qualified medical professionals for diagnosis and treatment."

NFR-7.2: User shall explicitly consent to data collection and privacy policy before account creation.

NFR-7.3: Users shall have right to download their data in CSV format and delete their account.

NFR-7.4: Doctor verification shall comply with PMC requirements before account activation.

NFR-7.5: Patient data shall be treated as confidential per Pakistan data protection regulations.

NFR-7.6: All communications with patients shall be HIPAA-aligned (confidentiality of medical records).

NFR-8.1: Web app shall work on modern browsers: Chrome (latest 2 versions), Firefox, Safari, Edge.

NFR-8.2: Mobile app shall support Android 8.0 and above.

NFR-8.3: System shall work on 3G networks; optimal on 4G/5G.

NFR-8.4: Responsive design shall adapt to screen sizes: 320px (mobile) to 2560px (desktop).

7. EXTERNAL INTERFACE REQUIREMENTS

7.1 Hardware Interfaces (EI-1)

EI-1.1 Smartphone Camera:

- Capture images of medical reports (prescriptions, lab reports, X-rays)
- Support both front and rear cameras
- Image resolution minimum 1080p for clarity

EI-1.2 Microphone & Speaker:

- Enable voice calls during video consultations
- Echo cancellation to reduce background noise
- Microphone gain adjustment for various noise environments

EI-1.3 GPS:

- Determine patient's location for doctor proximity matching
 - Accuracy within 100 meters
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7.2 Software Interfaces (EI-2)

EI-2.1 Agora SDK (Video Calling):

- Initialize video calls with low latency (<150ms)
- Support audio-only fallback if video fails
- Automatic bitrate adjustment based on connection quality

EI-2.2 Google Vision API (OCR):

- Extract text from medical report images

- Confidence scores for extracted text
- Fallback: Tesseract OCR for offline capability

EI-2.3 Firebase Cloud Messaging:

- Send push notifications for appointment reminders
- SMS gateway integration (Twilio or local provider)

EI-2.4 Google Translate API / Multilingual Models:

- Translate AI responses to Urdu
- Maintain medical terminology accuracy

EI-2.5 Payment Gateway APIs:

- JazzCash API for mobile money payments
- EasyPaisa API for wallet payments
- Handle payment confirmations and receipts

EI-2.6 Google Maps API:

- Display doctor locations on map
- Calculate distances between user and doctor
- Geocode addresses to coordinates

EI-2.7 Voice Agent APIs (VoIP & Speech):

- **Twilio Voice API:** Outbound calling to clinics
- **Speech-to-Text (STT):** Google Cloud Speech-to-Text or Azure Speech Services
- **Text-to-Speech (TTS):** Google Cloud Text-to-Speech or similar
- **Voice AI Conversation:** Custom voice agent logic (conversational flow, context understanding)
- Call recording, transcription, and logging
- Support for multiple languages (English and Urdu)

7.3 Communication Interfaces (EI-3)

EI-3.1 Real-Time Chat:

- WebSocket protocol for low-latency messaging

- Supports text, emoji, and timestamps
- Message delivery confirmation

EI-3.2 Video/Audio Calling:

- Agora SDK provides encrypted, peer-to-peer video
- Fallback to audio if bandwidth limited
- Call recording (with consent) for consultation records

EI-3.3 Notification Channels:

- Push notifications (Firebase)
- SMS notifications (Twilio / local SMS gateway)
- Email notifications (SendGrid or similar)

EI-3.4 API Security:

- All API requests use JWT authentication
 - HTTPS/TLS encryption for data in transit
 - Rate limiting per API endpoint
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8. SYSTEM CONSTRAINTS & ASSUMPTIONS

8.1 Assumptions

AS-1.1 Internet Connectivity:

- Users have consistent internet access (minimum 3G speed)
- Internet reliability in Pakistan shall remain stable during project duration

AS-1.2 Device Availability:

- Target patients have access to smartphones or computers
- Device has minimum 2GB RAM and sufficient storage

AS-1.3 User Technical Literacy:

- Patients can use basic mobile applications
- Voice support and simple UI accommodate low-literacy users

- Doctors are familiar with video conferencing

AS-1.4 Medical Data Quality:

- Medical reports uploaded are legible and in standard formats
- Users have access to their medical history records

AS-1.5 Doctor Availability:

- Licensed doctors from Pakistan will register and provide consultations
- Doctors maintain regular availability schedules

AS-1.6 AI Model Performance:

- Pre-trained medical NLP models (BioBERT, MedBERT) are available
- Models achieve acceptable accuracy after fine-tuning on available datasets

AS-1.7 Third-Party Service Stability:

- Agora, Firebase, payment gateways remain operational
- API uptime $\geq 99\%$

AS-1.8 Regulatory Compliance:

- Medical disclaimers are legally sufficient for MVP
- No clinical trial approval required for academic prototype

AS-1.9 User Trust:

- Users accept AI as preliminary guidance, not diagnosis
- Users are willing to share medical information securely
- Telemedicine is accepted in Pakistan's healthcare culture

8.2 Dependencies

DP-1: Third-Party Services

- Agora SDK for video calling (requires subscription)
- Google Vision API for OCR (pay-per-use)
- Firebase for real-time database and notifications
- JazzCash/EasyPaisa payment APIs

- Google Maps API for location services

DP-2: AI/ML Resources

- HuggingFace Transformers library (open-source)
- Pre-trained medical models (BioBERT, MedBERT from research repos)
- Medical knowledge database for symptom-to-specialty mapping
- Training dataset for fine-tuning (will use public medical datasets)

DP-3: Cloud Infrastructure

- AWS or Google Cloud Platform for hosting
- Auto-scaling to handle concurrent users
- Managed database services (RDS, Cloud SQL)

DP-4: External Data

- PMC doctor database for credential verification (may need manual verification for MVP)
- Standard medical reference ranges for vital signs and test values
- Symptom-to-specialty mapping database

DP-5: Development Tools

- React/Vue for frontend
- Node.js/Python for backend
- PostgreSQL for primary database
- Git for version control
- Docker for containerization

9. PHASE 2 FEATURES (Deferred - Not in Scope for 4-Month Timeline)

These features are planned for Phase 2 after MVP validation: